

Computer Application in

□ To provide the students with the theoretical knowledge to deal with the basics of

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3251 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3251.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Application and Business Management
Course code	3251
Course title	Computer Application in
Semester	3 Credits: 3
Course category	Foundation Course
Periods per week	3 (L:3, T:0, P:0) Periods per semester: 45

Item	Official information
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide the students with the theoretical knowledge to deal with the basics of computer hardware and software.
- To lay foundation in basic aspects like operating systems, data processing, data communication, computer networks etc which is very well required in the management of an organization.

Course outcomes

CO	Official outcome	Study level
CO1	from the evolution, basic architecture and 12 Understanding components Identify the different Operating Systems and their	See official syllabus
CO2	10 Applying functions Utilize the concepts of Data Processing and Data	See official syllabus
CO3	11 Applying Communication Interpret the basics of Computer Networks and	See official syllabus
CO4	10 Applying Internet Series Exam 2	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	architecture and components.	4	As specified
Module 2	Identify the different operating systems and their functions	2	As specified
Module 3	Utilize the concepts of Data Processing and Data Communication	4	As specified
Module 4	Interpret the basics of Computer Networks and Internet	3	As specified

MODULE 1

architecture and components.

This module is presented from the official syllabus scope for Computer Application in. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: architecture and components.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Remembering classification, and generations of computers. Understand the basic architecture of a computer,

3 Remembering classification, and generations of computers. Understand the basic architecture of a computer, is an official topic in Module 1 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering classification, and generations of computers. Understand the basic architecture of a computer, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering classification, and generations of computers. understand the basic architecture of a computer, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering classification, and generations of computers. Understand the basic architecture of a computer, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Remembering classification, and generations of computers. Understand the basic architecture of a computer, definition/meaning. application definition/meaning. steps, application definition/meaning. Technical terms English-Malayalam.

the different components of computer, computer 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of

the different components of computer, computer 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of is an official topic in Module 1 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify the different components of computer, computer 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, the different components of computer, computer 4 understanding memory and types of memory. identify the types of data representations in computers, hardware, software, and types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What the different components of computer, computer 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഈയ്ക്കു് ഡിഫറന്റ് കമ്പോണന്റ്സ് ഓഫ് കമ്പ്യൂട്ടർ, കമ്പ്യൂട്ടർ 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of ഡാറ്റാ റിപ്രസന്റേഷൻ. ഡാറ്റാ റിപ്രസന്റേഷൻ, steps, application ഡാറ്റാ റിപ്രസന്റേഷൻ ഡാറ്റാ. Technical terms English-ഈ ഡാറ്റാ റിപ്രസന്റേഷൻ.

software, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life

software, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life is an official topic in Module 1 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify software, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, software, and differentiate between system 2 understanding software and application software, free and open source software understand the software development life should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What software, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എസ് സോഫ്റ്റ്വെയർ, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life Cycle ഡെവലപ്മെന്റ് ലൈസൻസ് definition/meaning എസ്ഡിഎൽ. എസ്ഡിഎൽ ഫേസ് സ്റ്റേജുകൾ, steps, application ഏത് ടൂളുകളാണ് ഉപയോഗിക്കുന്നത്. Technical terms English-ഉം മലയാളത്തിലും.

3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart.

3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart. is an official topic in Module 1 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding cycle, algorithm and flow chart. introduction to computer systems introduction to computers, evolution of computers, characteristics of computers, classification of computers, generations; basic computer architecture; components of computer systems (i/o devices), computer memory and its types; data representation basics; computer software, relation between hardware and software; types of software - system software, application software; free and open source software, software development life cycle; introduction to algorithm and flow chart. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

architecture and components.

Know the evolution, characteristics,

M1.01

3

Remembering

classification, and generations of computers.

Understand the basic architecture of a computer,

M1.02

4

Understanding

memory and types of memory.

Identify the types of data representations in

computers, hardware, software, and types of

M1.03

2

Understanding

software, and differentiate between system

software and application Software, Free and open

source software

Understand the Software Development Life

M1.04

3

Understanding

Cycle, algorithm and flow chart.

Contents:

Introduction to Computer Systems

Introduction to computers, Evolution of computers, Characteristics of computers,

Classification of computers, Generations; Basic Computer Architecture; Components of

Computer Systems (I/O Devices), Computer Memory and its types; Data

Representation

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Identify the different operating systems and their functions

This module is presented from the official syllabus scope for Computer Application in. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the different operating systems and their functions
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

8 Understanding systems, and the techniques used.

8 Understanding systems, and the techniques used. is an official topic in Module 2 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding systems, and the techniques used. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding systems, and the techniques used. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding systems, and the techniques used. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-8 Understanding systems, and the techniques used. ഓരോ ഘട്ടത്തിനും definition/meaning നൽകണം. ഓരോ ഘട്ടത്തിനും steps, application നൽകണം. Technical terms English-ൽ നൽകണം.

Compare the different operating systems. 2 Applying Operating Systems
Introduction to Operating Systems, Definition of OS, Layers of OS; Types of
Operating Systems - single user & multi user OS, batch processing OS, Time
sharing OS, Real Time OS; Operating System Techniques - Multiprogramming,
Multitasking, Multithreading, Multiprocessing; Some Popular Operating
Systems - DOS (Disk Operating System), UNIX Operating System, Linux,
Microsoft Windows, Microsoft Windows NT, Mobile Operating systems

Compare the different operating systems. 2 Applying Operating Systems
Introduction to Operating Systems, Definition of OS, Layers of OS; Types of
Operating Systems - single user & multi user OS, batch processing OS, Time
sharing OS, Real Time OS; Operating System Techniques - Multiprogramming,
Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems -
DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows,
Microsoft Windows NT, Mobile Operating systems is an official topic in Module 2
of Computer Application in. Study the terminology first, then connect the stated
purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare the different operating systems. 2 Applying Operating Systems Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare the different operating systems. 2 applying operating systems introduction to operating systems, definition of os, layers of os; types of operating systems - single user & multi user os, batch processing os, time sharing os, real time os; operating system techniques - multiprogramming, multitasking, multithreading, multiprocessing; some popular operating systems - dos (disk operating system), unix operating system, linux, microsoft windows, microsoft windows nt, mobile operating systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare the different operating systems. 2 Applying Operating Systems Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Compare the different operating systems. 2 Applying Operating Systems Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems
 പരസ്പരം താരതമ്യം ചെയ്തുകൊണ്ട് operating systems-ന്റെ definition/meaning നൽകുക. operating systems-ന്റെ components, steps, application നൽകുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Identify the different operating systems and their functions

Explain the different functions, types of operating

M2.01

8

Understanding

systems, and the techniques used.

M2.02

Compare the different operating systems.

2

Applying

Series Test – I

1

Contents:

Operating Systems

Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating

Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time

OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Utilize the concepts of Data Processing and Data Communication

This module is presented from the official syllabus scope for Computer Application in. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Utilize the concepts of Data Processing and Data Communication
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain data and file processing methods 3 Understanding Describe the file organization methods and file

Explain data and file processing methods 3 Understanding Describe the file organization methods and file is an official topic in Module 3 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain data and file processing methods 3 Understanding Describe the file organization methods and file using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain data and file processing methods 3 understanding describe the file organization methods and file should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain data and file processing methods 3 Understanding Describe the file organization methods and file means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓരോ Explain data and file processing methods 3 Understanding Describe the file organization methods and file ഓരോരോരോരോരോരോരോ definition/meaning ഓരോരോരോരോരോരോരോ. ഓരോരോരോ ഓരോരോരോരോ, steps, application ഓരോരോരോ ഓരോരോരോരോരോരോരോ. Technical terms English-ഓരോരോരോ ഓരോരോരോരോരോരോരോ.

3 Understanding operations Identify the importance of database management

3 Understanding operations Identify the importance of database management is an official topic in Module 3 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding operations Identify the importance of database management using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding operations identify the importance of database management should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding operations Identify the importance of database management means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 3 Understanding operations Identify the importance of database management ഡാറ്റാബേസ് മാനേജ്മെന്റ് definition/meaning ഡാറ്റാബേസ് മാനേജ്മെന്റ്. ഡാറ്റാബേസ് മാനേജ്മെന്റ് steps, application ഡാറ്റാബേസ് മാനേജ്മെന്റ്. Technical terms English- ഡാറ്റാബേസ് മാനേജ്മെന്റ്.

1 Understanding system Recognize the data communication modes,

1 Understanding system Recognize the data communication modes, is an official topic in Module 3 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding system Recognize the data communication modes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding system recognize the data communication modes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding system Recognize the data communication modes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 1 Understanding system Recognize the data communication modes, definition/meaning .
 , steps, application . Technical terms English- .

transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems.

transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems. is an official topic in Module 3 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transmission methods and types of 4 applying communication media. business data processing and data communication data processing; file management system, types of processing files; file organization; file operations;

database management system, its purpose and uses. basic elements of a communication system; modes of data communication; transmission basics - analog and digital; types of data transmission media; modems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems.

പ്രധാനപ്പെട്ട പദങ്ങൾക്ക് definition/meaning നൽകണം. പദങ്ങൾക്ക് പദങ്ങൾ, steps, application നൽകണം. Technical terms English-ൽ പദങ്ങൾ നൽകണം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Utilize the concepts of Data Processing and Data Communication

M3.01	Explain data and file processing methods	3	Understanding
M3.02	Describe the file organization methods and file operations	3	Understanding
M3.03	Identify the importance of database management system	1	Understanding
M3.04	Recognize the data communication modes, transmission methods and types of communication media.	4	Applying

Contents:

Business Data Processing and Data Communication

Data Processing; File Management System, Types of processing files; File Organization;

File operations; Database Management System, its purpose and uses.

Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Interpret the basics of Computer Networks and Internet

This module is presented from the official syllabus scope for Computer Application in. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Interpret the basics of Computer Networks and Internet
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the different types of computer networks

Explain the different types of computer networks is an official topic in Module 4 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the different types of computer networks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the different types of computer networks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the different types of computer networks means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain the different types of computer networks. Define/meaning. Explain the steps, application of computer networks. Technical terms English-Malayalam.

Identify the different network topologies

Identify the different network topologies is an official topic in Module 4 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the different network topologies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the different network topologies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the different network topologies means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-☐☐ Identify the different network topologies ☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐ ☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/>

Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/> is an official topic in Module 4 of Computer Application in. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe about the basics of internet. 3 understanding computer networks need and definition for computer communication networks; advantages of networks; types of network; network topologies; internet basics - isp, web server, web browser, domain names, world wide web, e-mail, search engines. text / reference: t/r book title/author r1 rajaraman, fundamentals of computers, prentice hall of india, 2003 p.k sinha, priti sinha, computer fundamentals, kalyani publishers, 2nd r2 edition r3 shelly, business data communication, thomson learning, bombay, 2003 white, data communication & computer network, thomson learning, r4 bombay, 2012 r5 b. ram, computer fundamentals, new age international, 2000 online resources: sl.no website link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals

Aspect	What to prepare
	of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 https://www.javatpoint.com/computer-fundamentals-tutorial 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 http://examradar.com/communication-networking-summary-1/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/> definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Interpret the basics of Computer Networks and Internet

M4.01	Explain the different types of computer networks	4	
	Understanding		
M4.02	Identify the different network topologies	3	Applying
M4.03	Describe about the basics of Internet.	3	
	Understanding		
	Series Test – II	1	

Contents:

Computer Networks

Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines.

Text / Reference:

T/R	Book Title/Author
R1	Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003
	P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd
R2	Edition
R3	Shelly, Business Data Communication, Thomson Learning, Bombay, 2003

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Remembering classification, and generations of computers. Understand the basic architecture of a computer,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering classification, and generations of computers. Understand the basic architecture of a computer,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain the different components of computer, computer 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of. (3 marks)

Answer: Begin with the standard definition or identity of *the different components of computer, computer 4 Understanding memory and types of memory. Identify the types of data representations in computers, hardware, software, and types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain software, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life. (3 marks)

Answer: Begin with the standard definition or identity of *software, and differentiate between system 2 Understanding software and application Software, Free and open source software Understand the Software Development Life*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source

software, Software Development Life Cycle; Introduction to Algorithm and Flow chart.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Cycle, algorithm and flow chart. Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 8 Understanding systems, and the techniques used.. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding systems, and the techniques used..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Compare the different operating systems. 2 Applying Operating Systems Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems. (3 marks)

Answer: Begin with the standard definition or identity of *Compare the different operating systems*. 2 *Applying Operating Systems Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Explain data and file processing methods 3 Understanding Describe the file organization methods and file. (3 marks)

Answer: Begin with the standard definition or identity of *Explain data and file processing methods 3 Understanding Describe the file organization methods and file*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding operations Identify the importance of database management. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding operations Identify the importance of database management*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain 1 Understanding system Recognize the data communication modes,. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding system Recognize the data communication modes,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems.. (3 marks)

Answer: Begin with the standard definition or identity of *transmission methods and types of 4 Applying communication media. Business Data Processing and Data Communication Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses. Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□.

Module 4

Define or explain Explain the different types of computer networks. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the different types of computer networks*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify the different network topologies. (3 marks)

Answer: Begin with the standard definition or identity of *Identify the different network topologies*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/>. (3 marks)

Answer: Begin with the standard definition or identity of *Describe about the basics of Internet. 3 Understanding Computer Networks Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide*

Web, E-mail, Search Engines. Text / Reference: T/R Book Title/Author R1 Rajaraman, Fundamentals of Computers, Prentice Hall of India, 2003 P.K Sinha, Priti Sinha, Computer Fundamentals, Kalyani Publishers, 2nd R2 Edition R3 Shelly, Business Data Communication, Thomson Learning, Bombay, 2003 White, Data Communication & Computer Network, Thomson Learning, R4 Bombay, 2012 R5 B. Ram, Computer Fundamentals, New Age International, 2000 Online Resources: Sl.No Website Link 1 <https://www.javatpoint.com/computer-fundamentals-tutorial> 2 https://www.tutorialspoint.com/computer_fundamentals/index.htm 3 https://swayam.gov.in/nd2_cec19_cs06/preview 4 <http://examradar.com/communication-networking-summary-1/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Remembering classification, and generations of computers. Understand the basic architecture of a computer,**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **8 Understanding systems, and the techniques used.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain data and file processing methods 3 Understanding Describe the file organization methods and file.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain the different types of computer networks.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.javatpoint.com/computer-fundamentals-tutorial>
- https://www.tutorialspoint.com/computer_fundamentals/index.htm
https://swayam.gov.in/nd2_cec19_cs06/preview
- <http://examradar.com/communication-networking-summary-1/>

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 3251. Verify institutional updates against the current official curriculum.